CHAPTER 4

Logic and Control

Programs, even simple ones, will inevitably have to alter their behavior based on their inputs, or current state. Likewise, it's usually necessary to perform operations repeatedly: imperative languages (programming languages that are composed of step-by-step instructions) such as PHP have features that allow for the testing of data and the concise expression of repetitive sequences of code. This chapter explores logical operations and control structures. Logic operations generate boolean values (`true` or `false`), and the control structures of PHP often use these boolean values to determine their behavior. In this way we can build programs that make runtime decisions and achieve a great deal of work with minimal code.

PHP, like most languages based on C syntax, has the usual language constructs for controlling the flow of execution. Logic branches with `if` statements, compound variables can be iterated with `for` and `foreach`, and code blocks can be looped with `do` and `while`. More advanced forms of logic branching are available with `switch` and the recently introduced `match` construct.

In Chapter 1, "Getting Started," I introduced the concept of expressions. These are a high-level aspect of the syntax of PHP (indeed, of any programming language). Expressions consist of sequences of operations that are evaluated (generate data values), and they can be thought of as "code that reduces to a value." An expression, however complex, will always evaluate – or reduce – to a value. That value can then be tested – and the testing itself is an expression too. Chapter 1, "Getting Started," showed that expressions could be numeric. `1 + 1` reducing to `2` is a simple example. String operations are also expressions, like `'Hello' . ' world'` reduces to the single string `'Hello world'`. They can be more complex, of course, consisting of function calls, array access, or any other legal PHP operation.

A logical expression is one that ultimately evaluates to `true` or `false`. They usually contain comparison and/or logical operators, which are used to test operands, where the operands can be variables, literal values, or sub-expressions. Therefore, the key

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